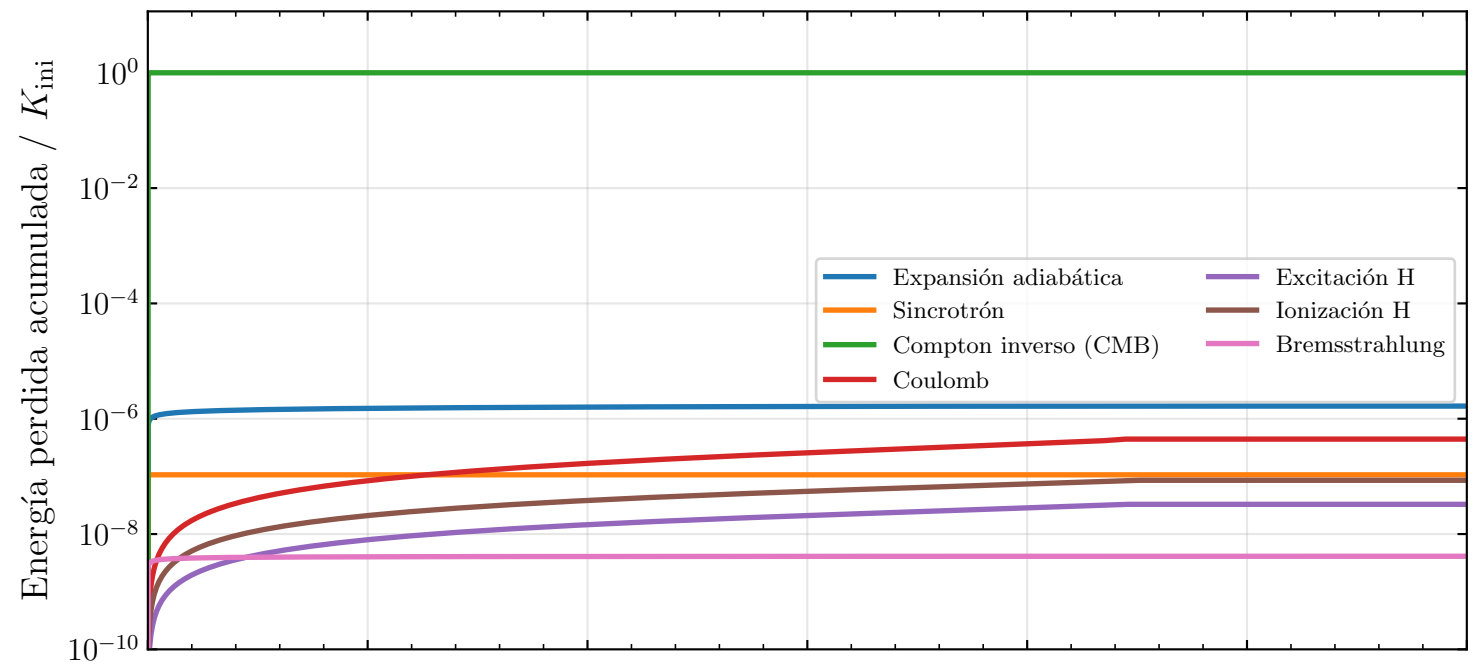
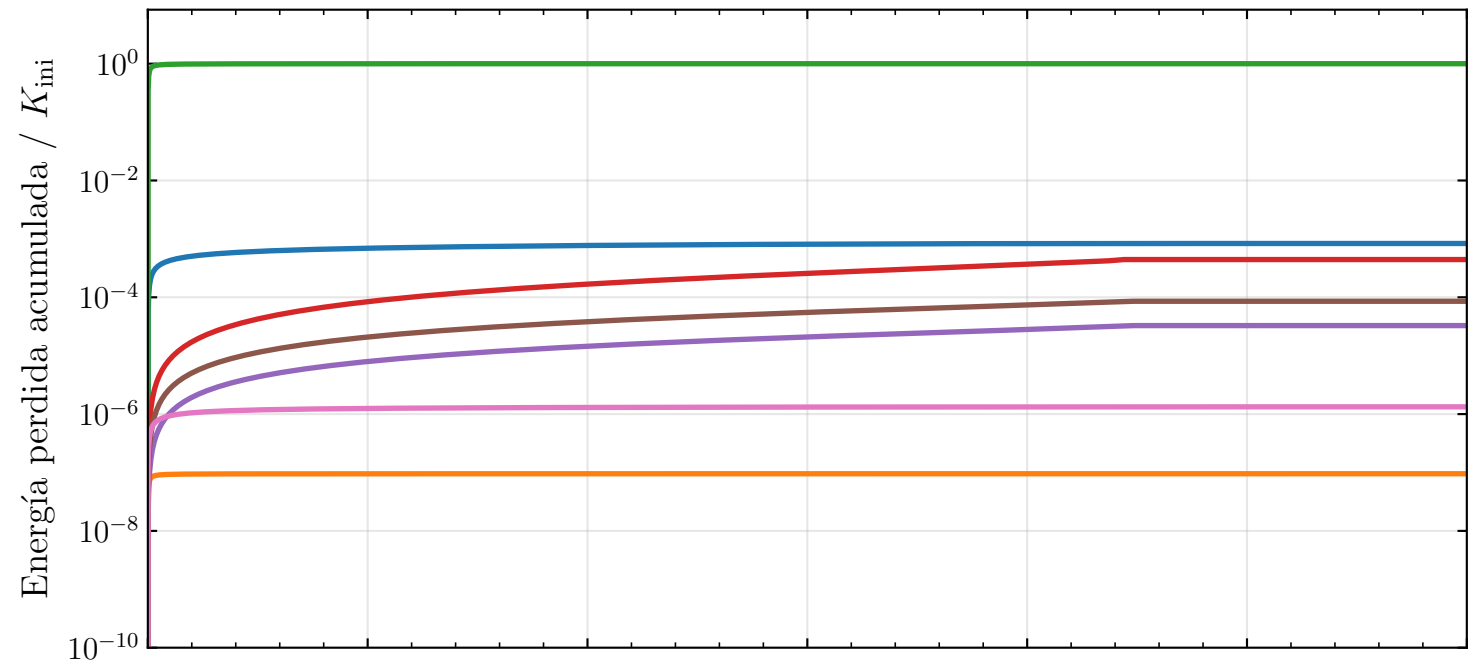


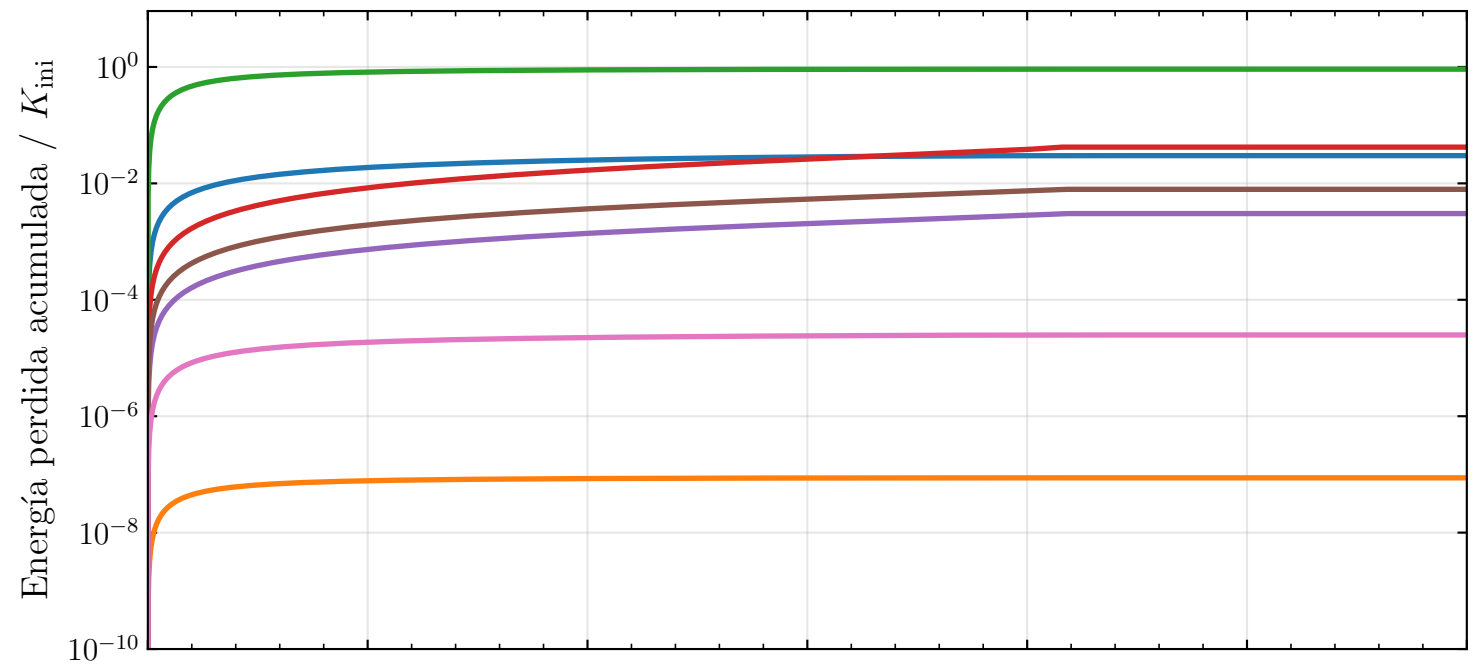
$x_e = 0.5$
 $K_{\text{ini}} = 10^{12} \text{ eV}$



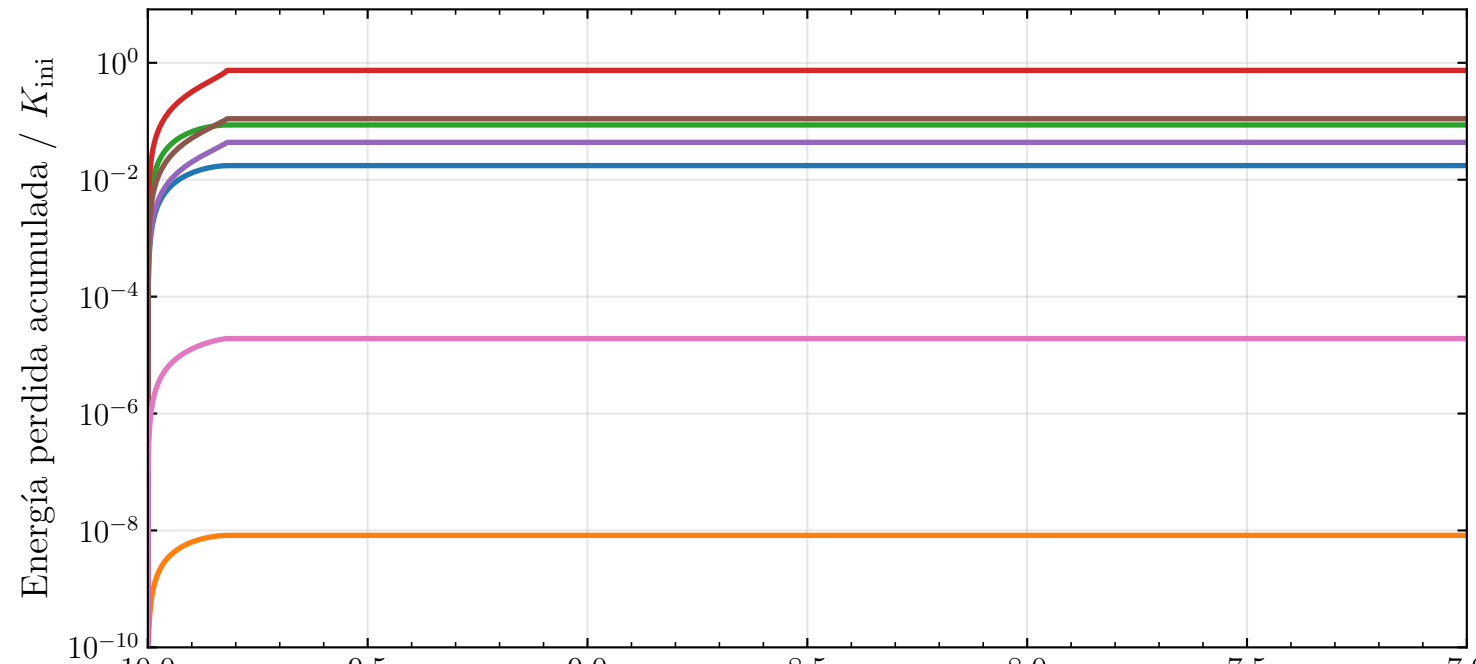
$K_{\text{ini}} = 10^9 \text{ eV}$



$K_{\text{ini}} = 10^7 \text{ eV}$



$K_{\text{ini}} = 10^5 \text{ eV}$



Redshift z